Dynamic Digital Twin Clinical Report

Patient ID: **P02194** | Generated for Rehabilitation Tracking

1. Patient Profile & Clinical Baseline

Age:	29	**Gender:**	Male
Injury Type:	ACL Tear	**Severity:**	Severe
Surgery Required:	No	**BMI:**	25.5
Activity Level:	Sedentary	**Smoking Status:**	Never

2. Twin Recovery State — Week 1

Metric	Recorded Value	Status / Target
Pain Score	6.3 / 10	Target <= 2.0
Range of Motion (ROM)	58.2°	Target >= 120°
Quadriceps Strength	36.8%	Target >= 80%
Hamstring Strength	30.0%	Target >= 80%
Balance Score	36.0	Target >= 75
Walking Speed	0.49 m/s	Target >= 1.2 m/s
Swelling	Severe	Target: None
Compliance	83.9%	Target >= 80%

3. Pressure Points & Flagged Recovery Concerns

✓ No critical recovery flags detected for Week 1

5. Treatment Pathway Clones & Reaction Analysis

Pathway Option	Est. Duration	Predicted Recovery	Reaction Profile
Clone A: Standard Physiotherapy	40 weeks	94%	Steady baseline progression. Moderate joint flare-up risk in mid-phase.
Clone B: Hydrotherapy & Accelerated Track	32 weeks	99%	Rapid pain relief and accelerated early ROM gain due to reduced weight-bearing stress.
Clone C: Cautious Balance & Low-Impact	46 weeks	89%	Ultra-low strain protocol. Slower timeline but near-zero risk of swelling or re-injury.